Final Exam - EEE3094S 2020 - MEMO

Information

- The exam is open book.
- There are seven questions, totalling 100 marks. You must answer all of them.
- Submit your answers in pdf format. They may be typed or handwritten. Sketches may be hand-drawn and scanned, or constructed in a program such as PowerPoint. A .pptx file containing the plots has been provided.
- You must explain your reasoning and show your calculations for all questions.
- If you create or modify a plot to support your answer, ensure that you reference this graphic in your answer. You should also label the graphic with the question number. It is also helpful to annotate important features to draw attention to them.
- Closed-loop systems are in unity feedback unless otherwise stated.

Question 1

Figure 1.A shows the block diagram of a system that can be represented by the transfer function

$$T(s) = \frac{65(s+2)}{s^4 + 7s^3 + 32s^2 + 132s + 221}$$

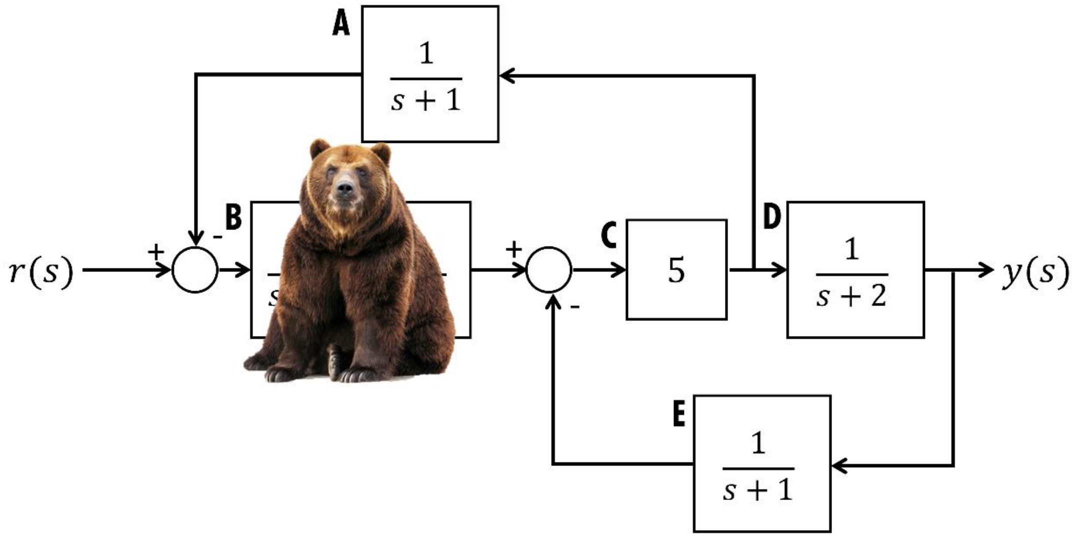


Figure 1.A – Block diagram.

A) [9 marks] Unfortunately, a bear is sitting in front of one of the blocks and refuses to move. You would prefer to resolve the situation peacefully, and luckily, your knowledge of block diagram algebra should allow you to do so. Use it to show that the transfer function of the obscured block must be $B(s) = \frac{13}{s^2+4s+13}$

Solution: the following steps can be used to simplify the block diagram to one block:

1) Move the take-off point between C and D to the output. A $\frac{1}{D}$ block must be included in the same branch as A.
2) Move the summing junction between B and C to combine with the summing junction at the input. A $\frac{1}{B}$ block must be included in the same branch as E.
3) Combine the cascade connections, to give $\frac{A}{D}$ in the top branch, BCD in the middle branch and $\frac{E}{B}$ in the bottom branch.
4) Combine the parallel connection to give $\frac{A}{D} + \frac{E}{B}$ in the feedback path
5) There should now be one block $G = BCD$ in the forward path and one block $H = \frac{A}{D} + \frac{E}{B}$ in the feedback path. These can be combined using the loop rule, $\frac{G}{1+GH}$, to give $\frac{BCD}{1+ABC+CDE}$

If the transfer functions of each block, and suggested transfer function for B are substituted into this function for the system, the result will be the given system transfer function.

B) [6 marks] The unit step response in Figure 1.B (on the next page) was produced by the hidden subsystem. With reference to the final value, overshoot, and settling time, show that it supports the model of $B(s)$ you worked out in part A.

Solution: based on the second-order transfer function form $\frac{A\omega_n^2}{s^2+2\zeta\omega_n+\omega_n^2}$, the damping ratio, natural frequency and gain A can be calculated to be $\zeta = 0.55$, $\omega_n = 3.6$ rad/s and A = 1

The final value of the given step response is 1, which confirms the value of A.

The percentage overshoot can be calculated to be 12.3% using the given formula, which agrees with the value of the peak overshoot shown in the plot.

Students can confirm the natural frequency using either the peak time formula, or calculate the real part of the pole and show that it gives a matching 2% or 5% settling time.

You may find the following formulas useful:

Percentage overshoot: $100e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}}$ Peak time: $\frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$

where ζ is the damping ratio and ω_n is the natural frequency.

Total: 15 marks

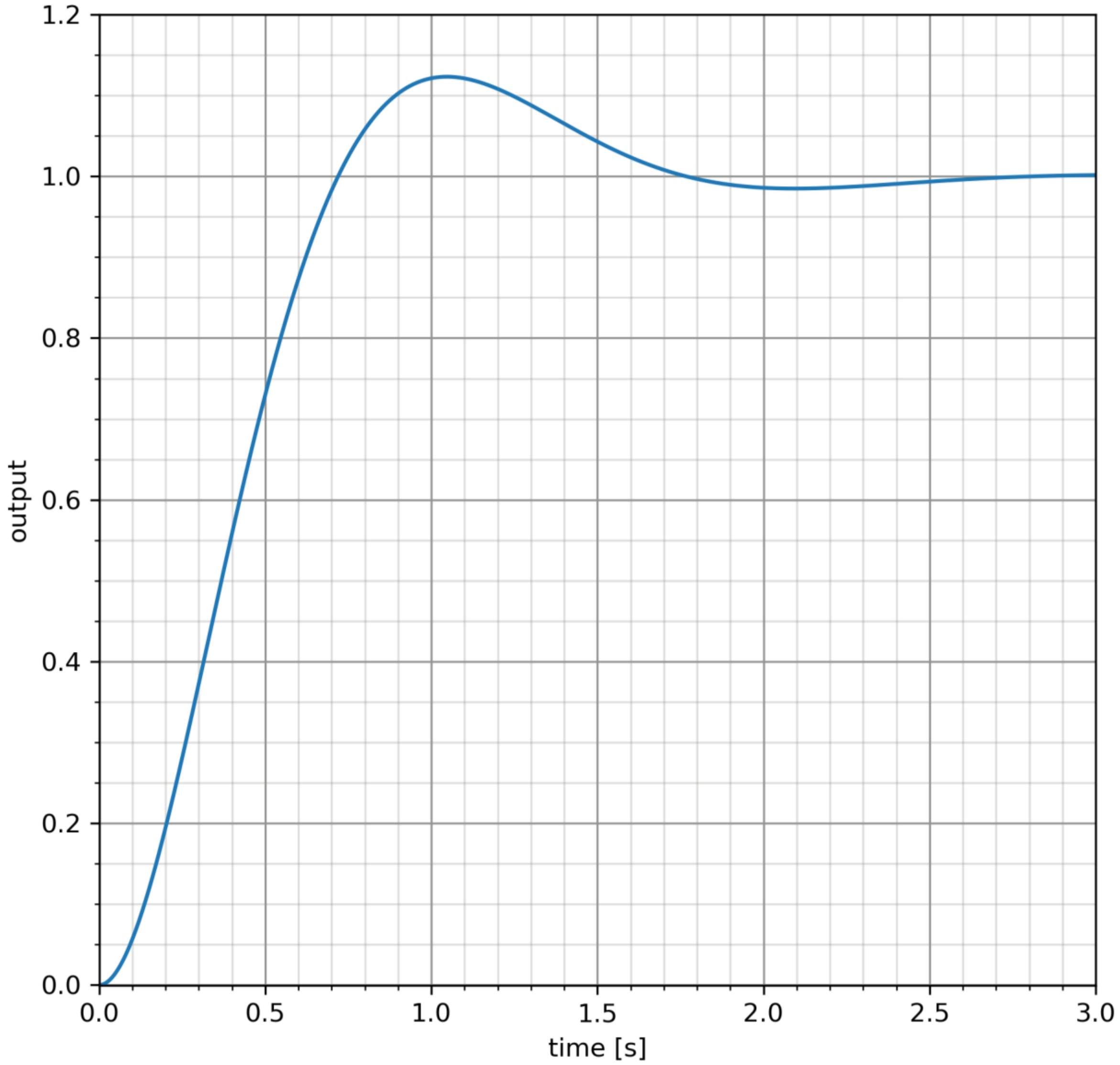


Figure 1.B – step response of $B(s)$

Question 2

Figure 2.A shows the block diagrams of three systems with the same type number.

A) [1 mark] What is the type number of these systems?

Solution: All three systems are type 1, as they have one integrator in the loop path.

B) For each of the following design specifications, explain whether it can be met by all, none or some of the given systems.

i) [2 marks] Tracking a position setpoint without error

Solution: all three systems can track a position setpoint without error, as the type number is equal to the order of the input.

ii) [2 marks] Tracking a ramp setpoint without error

Solution: none of the systems can track a ramp setpoint without error, as the type number is lower than the order of the input.

iii) [2 marks] Complete rejection of input disturbances in the form of a step

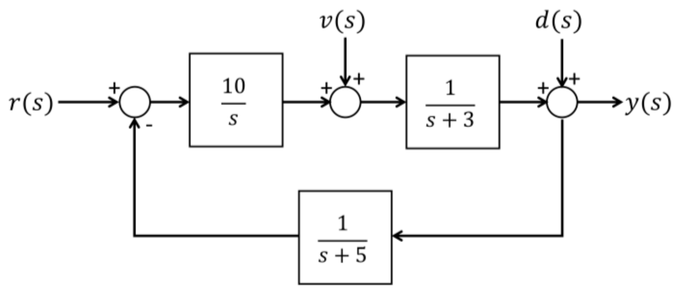
Solution: Systems A and C can reject an input disturbance completely. The transfer function relating the input disturbance to the output is $T_v(s) = \frac{G(s)}{1+K(s)G(s)H(s)}$ where $G(s), K(s)$ and $H(s)$ are the plant, controller and feedback path transfer functions, respectively. The DC gain of the controller or feedback path must therefore tend to infinity for this disturbance to tend to zero, which means that the integrating pole must be in one of these blocks to fulfil the requirement.

iv) [2 marks] Complete rejection of output disturbances in the form of a step

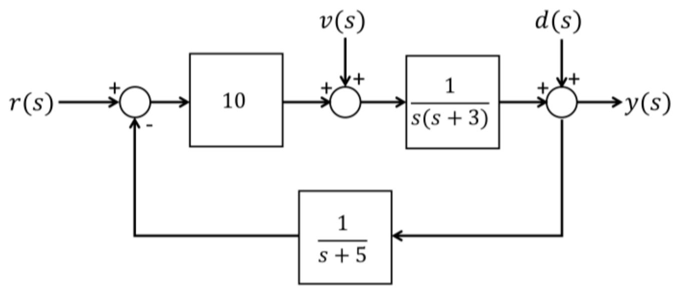
Solution: All three systems can reject an output disturbance completely. The transfer function relating the output disturbance to the output is $T_d(s) = \frac{1}{1+L(S)}$, where $L(s)$ is the transfer function of the loop path in cascade, so the disturbance will tend to zero if any of the systems in this path contain an integrator. (This is the same transfer function relating the error to the setpoint.)

Total: 9 marks

System A



System B



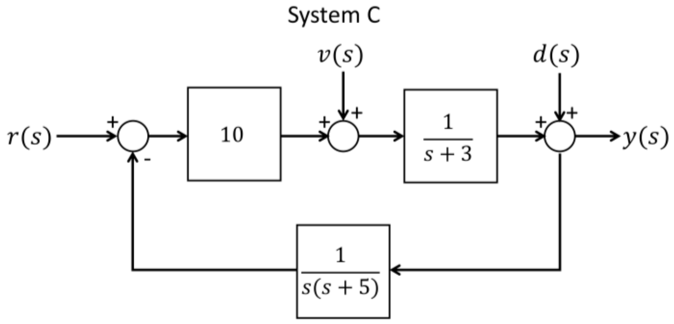


Figure 2.A – block diagrams of three systems. The setpoint, input disturbance and output disturbance are represented by the inputs $r(s)$, $v(s)$ and $d(s)$, respectively.

Question 3

Decide whether each of the following statements are TRUE or FALSE. Explain your reasoning.

A) [3 marks] Consider the inverse Nichols chart shown in figure 3.A. If this system experiences an output disturbance, frequency components of the disturbance close to 0.75 rad/s will be magnified in the output of the closed-loop system.

Solution: TRUE. The frequency response intersects the M circle representing a sensitivity of 2 decibels at 0.75 rad/s. Since the sensitivity has a magnitude greater than one, components of an output disturbance will be amplified.

B) [3 marks] The controller $C(s) = \frac{10(s-3)}{(s+5)}$ is a viable way to stabilize the unstable system $G(s) = \frac{1}{s-3}$ in closed loop.

Solution: FALSE. This controller attempts to cancel the unstable pole with a zero in the right-half plane. This is not a reliable way to stabilize the system in practise, as even a small error in the zero or pole position would allow the instability to persist.

C) [3 marks] The controller $C(s) = \frac{10(s+2)}{(s+10)}$ is a viable way to stabilize the unstable system $G(s) = \frac{1}{s-3}$ in closed loop.

Solution: FALSE. The gain of the controller is not high enough to draw the unstable pole into the left-half plane.

D) [3 marks] The discrete-time system $G(z) = \frac{12z}{z - 0.67}$ is stable.

Solution: TRUE. The system's pole is within the unit circle – the stable region of the z plane.

E) [3 marks] The model $G_r(s) = \frac{1}{(s+2)(s+3)}$ provides a reasonably accurate reduced-order approximation of the third-order system $G(s) = \frac{1}{(s+2)(s+3)(s+15)}$.

Solution: FALSE. Although the pole at -15 is five times faster than the other poles in the system, and therefore much less dominant, it has not been replaced by an equivalent gain in the proposed model, so the final value will not be the same as the original model.

Total: 15 marks

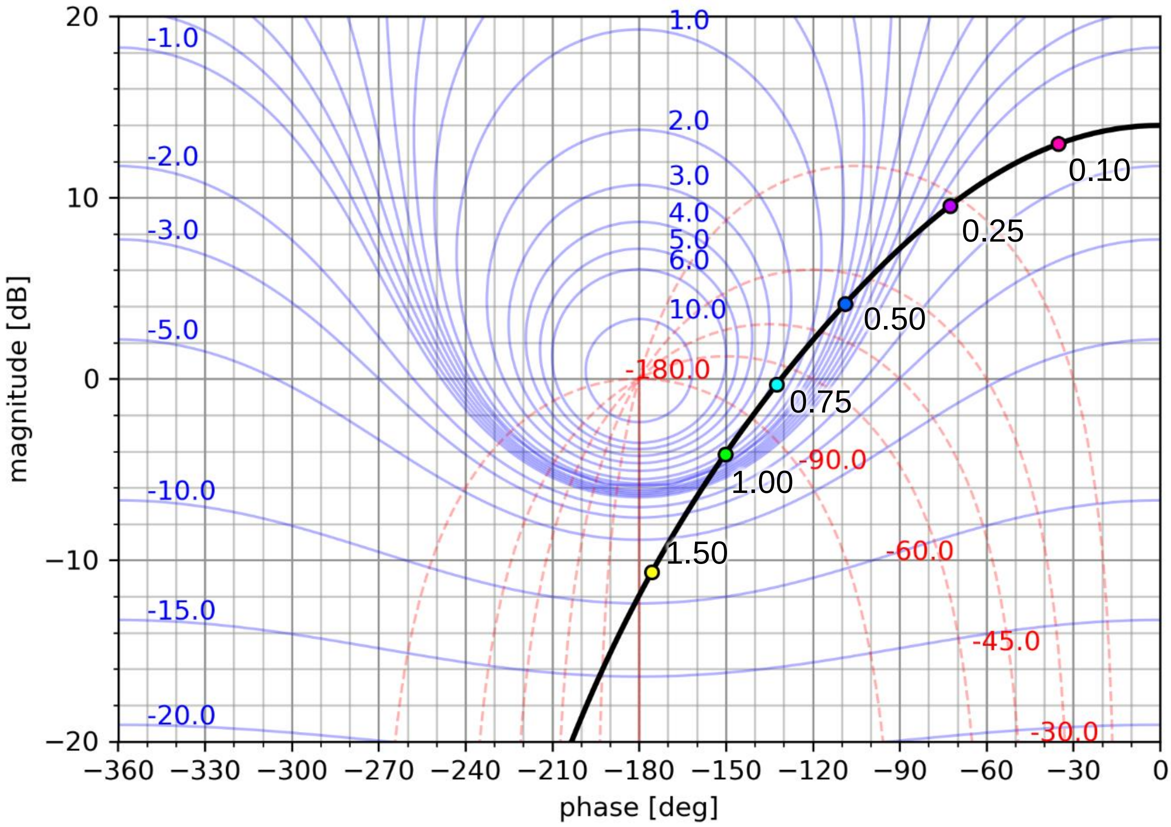
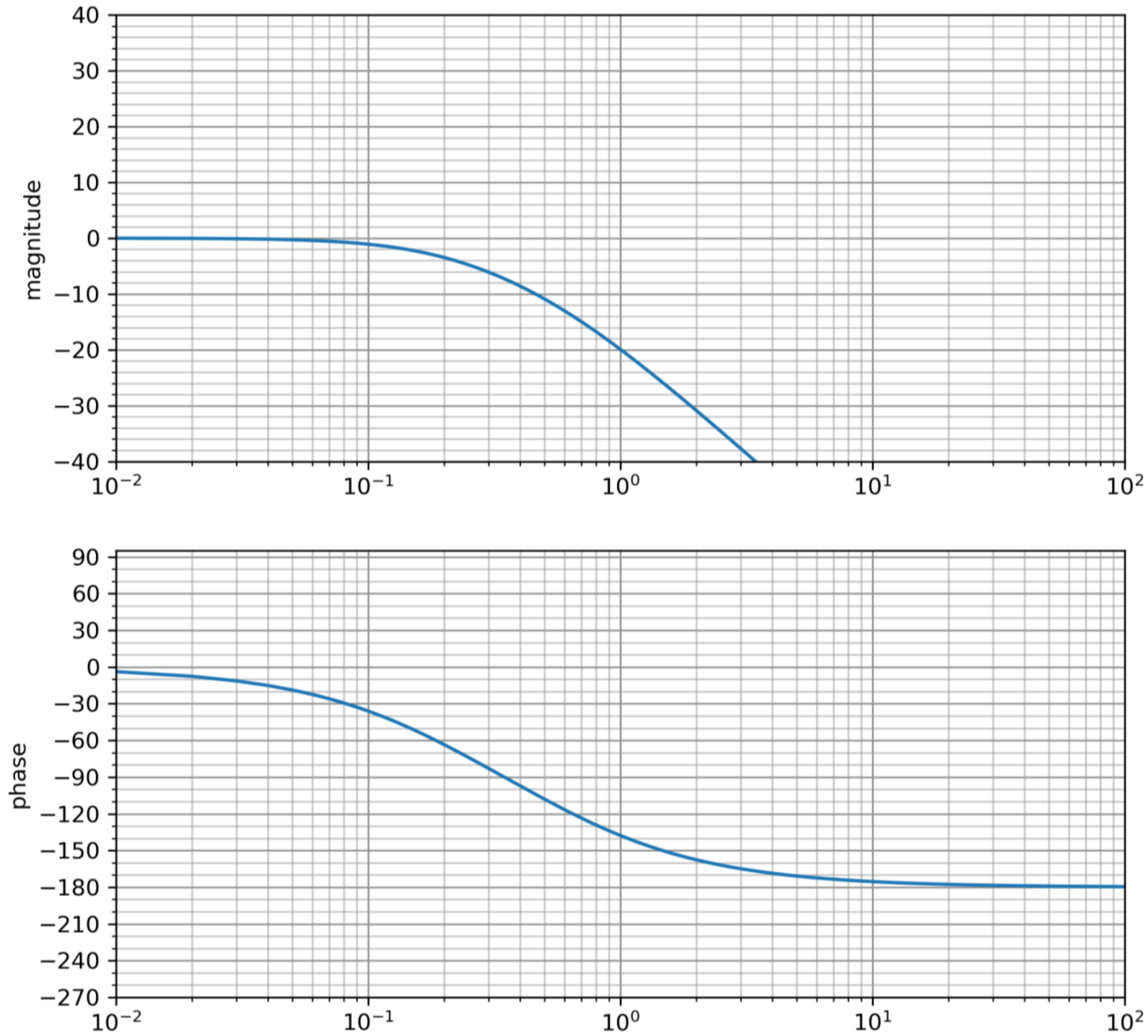


Figure 3.A – inverse Nichols chart. Annotated frequencies are in rad/s.

Question 4

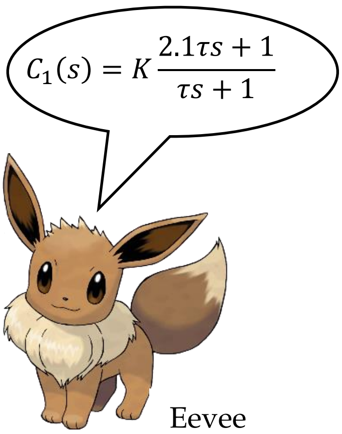
Figure 4.A shows the frequency response of a system G(s).



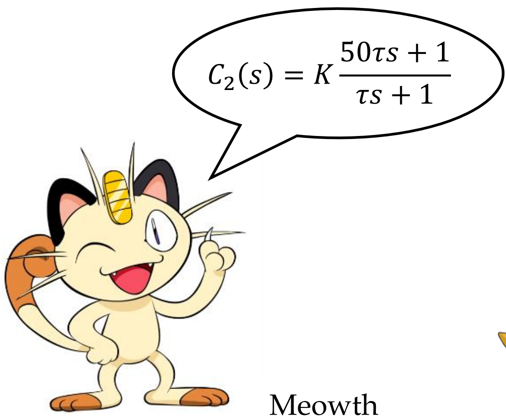
You are the head control engineer of a team that has been asked to design a lead compensator for this system. The requirements the design must satisfy include:

- Tracking a position input with >90% accuracy
- A phase margin of at least 60 degrees, because it is likely that the final controller will be implemented digitally.

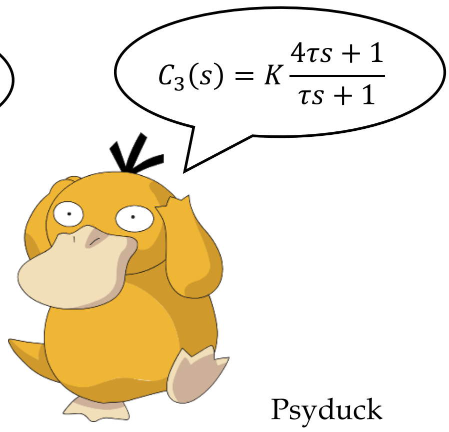
Your colleagues present the following partially-completed designs for you to evaluate:



Eevee



Meowth



Psyduck

A) [1 mark] Why is it necessary to increase the phase margin before implementing the controller digitally?

Solution: Like a time delay, sampling a system decreases the phase of its response, especially at high frequencies, so it is advisable to allow a large phase margin to prevent this from causing instability.

B)

i) [2 marks] Show that a gain of $K = 10$ meets the accuracy specification and sketch the Bode plot of the system at this gain on the template in Figure 4.B.

Solution: If the open loop gain $K = 10$, the closed-loop gain will be $A = \frac{10}{1+10} = 0.9$. The response will therefore have a final value exceeding 90% of a given position input. The response is plotted in figure 4.B.

ii) [1 mark] What is the phase margin of the system at this gain?

Solution: From the Bode plot sketched in the previous question, the phase margin is around 40 degrees.

C) [4 marks] Calculate the maximum phase that each design adds. Do you think that all the designs will be able to meet the phase margin specification?

Solution: The maximum phase can be calculated using the formula $\phi_p = \arcsin\left(\frac{\alpha-1}{\alpha+1}\right)$, where ϕ_p is the peak phase and α is the ratio of the pole position to the zero position. The values of α and ϕ_p for each controller are:

Eevee: $\alpha = 2.1, \phi_p = 20.8$ degrees

Meowth: $\alpha = 50, \phi_p = 73.9$ degrees

Psyduck: $\alpha = 4,\ \phi_p = 36.9$ degrees

Although all the designs provide enough additional phase to reach 60 degrees of phase margin at the peak, Eevee's design might not allow sufficient extra phase to compensate for the change in gain crossover frequency that occurs once the gain of the compensator's pole and zero are taken into account.

D) [2 marks] Meowth's design might be difficult to implement using analogue components. Explain why, and suggest a modification to fix the problem that would still provide the same phase lead.

Solution: The pole is much more than 10 times further away from the origin than the zero is. Implementing a conroller with a large ratio like this requires large values of the passive components used to build it. If this much additional phase is required, it is advisable to use two lead compensators with smaller α values in cascade instead.

E) Psyduck is confused about how to proceed.

i) [5 marks] Help him complete his design by calculating a suitable value of τ, and hence, the positions of the pole and zero of the compensator.

Solution: The compensator has a magnitude of $10\log_{10} 4 = 6$ decibels at the frequency where the peak phase lead occurs, so it should be aligned with the frequency where the system has a magnitude of -6 decibels. From the Bode plot in part B.i, this will be somewhere in the vicinity of 1.5 rad/s.

The time constant τ of the pole can be found using the formula $\tau = \frac{1}{\omega_p\sqrt{\alpha}} = \frac{1}{1.5\sqrt{4}} = 0.033$, which puts the pole at 30 and zero at 7.5

ii) [5 marks] Sketch an approximate Bode plot for the compensator design, and combined system on the same plot. You may add your sketches to Figure 5.B, or use the empty template in 5.B. Confirm whether the design satisfies the phase margin specification.

Solution: the plots are added to figure 4.B. Students may sketch a straight-line approximation of the bode plot. The rough shape of the combined system can be found by adding the responses at chosen points.

Total: 20 marks

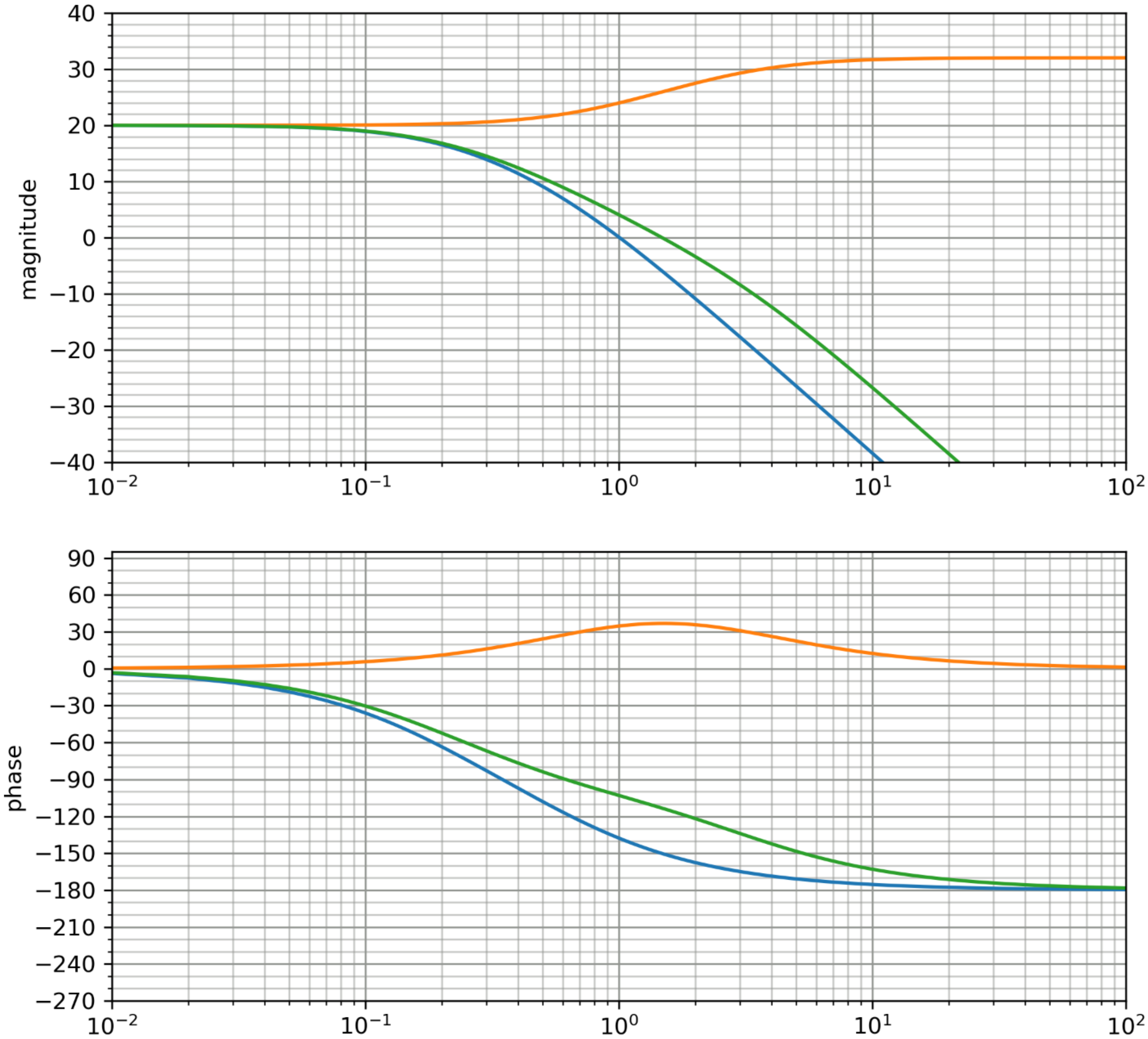


Figure 4.B – Bode plot template.

Memo: the response of the system with the gain increased by 20 dB (part B) is shown in blue.

The response of Psyduck's compensator (part E.ii.) is shown in orange, and the combined response of the response and system is shown in green.

Based on this response, the phase margin is now around 70 degrees, so it exceeds the requirement.

Question 5:

Figures 5A-C depict the frequency response of a system which is stable in open loop. The system can be approximated as second-order.

A) Based on this plot, approximate the system's
   i) [2 marks] gain margin

Solution: the response intersects the real axis at -0.3, so it will pass -1 and become unstable at a gain of 1/0.3 = 3.33

   ii) [2 marks] phase margin

Solution: the response intersects the unit circle at -0.8 - j0.6, which is arctan $\left(\frac{0.6}{0.8}\right)$ = 36.8 degrees away from the negative real axis.

   iii) [2 marks] accuracy for tracking a step setpoint in closed loop (as a percentage)

Solution: the DC gain of the response is 5, given by the point where it intersects the positive real axis. The closed-loop gain is therefore $A = \frac{5}{1+5}$ = 0.83. This equates to a position tracking accuracy of 83%.

   iv) [3 marks] damping ratio

Solution: the damping ratio can be approximated from the formula $\zeta = \frac{A}{100}\phi_m = \frac{0.83}{100}36.8 = 0.3$

   v) [6 marks] settling time to within 5% of its final value in closed loop

Solution: For a second-order system, the magnitude at the natural frequency will be $\frac{A}{2\zeta} = \frac{0.83}{2(0.3)} = 1.36$, or 2.6 decibels.

From the detailed Nyquist plot, we see that the marker representing 2.2 rad/s is just to the side of the M circle representing 2.5 decibels, so we can assume that this value is close to the natural frequency.

The real part of the pole is given by $\omega_n\zeta = (2.2)(0.3) = 0.66$, meaning the time constant is $\tau = \frac{1}{0.66} = 1.51$, so the response will settle to within 5% of its final value in approximately $3\tau = 3(1.51) = 4.5$ seconds.

B) This system is going to be used in an application where it will be subjected to a time delay of 0.2 seconds.
   i) [4 marks] Do you expect the closed-loop system to remain stable under these conditions?

Solution: the response intersects the unit circle at 1.6 radians per second, meaning the delay will produce a phase shift of $\omega T = (1.6)(0.2) = 0.32$ rad = 18.3 degrees at this frequency. This is smaller than the phase margin, so the system will not be destabilized.

   ii) [3 marks] What type of controller could you use to make the system more robust to delays? (You do not need to design the controller, just explain which of the controllers you have learned about would be suitable.)

Solution: an argument can be made for either the lead or lag compensators:

Lead compensator: can add lead to increase the phase margin

Lag compensator: can increase the phase margin indirectly by decreasing the low-frequency gain such that the gain crossover coincides with a phase that is further from -180 degrees.

Total: 20 marks

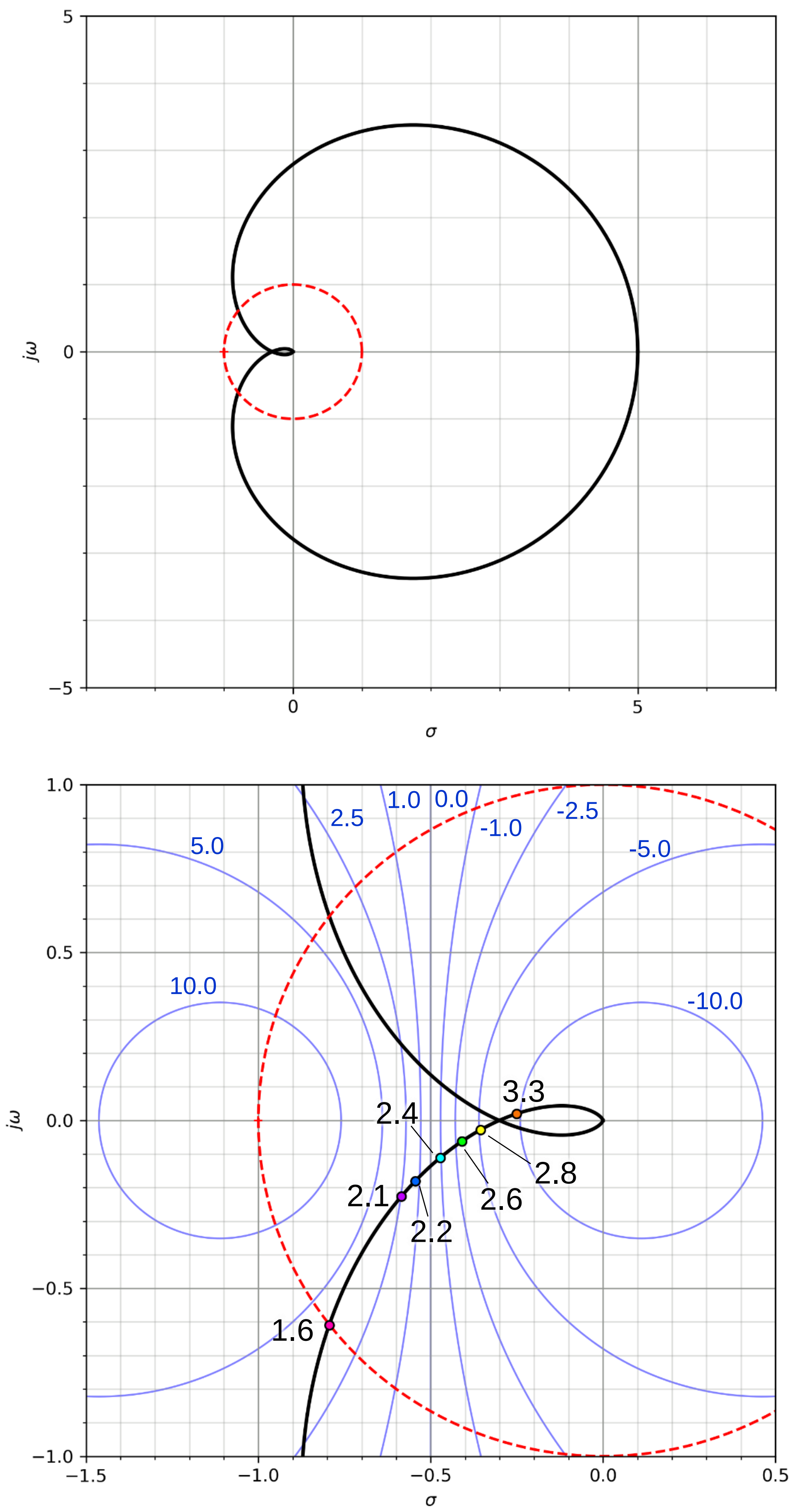


Figure 5.A Nyquist plot of system, and detailed view of Nyquist plot.

M circles are shown in blue. Their magnitudes are given in decibels.

The unit circle is shown in red.

Annotated frequencies are in rad/s

Question 6:

Figure 6.A shows the unit step responses of three systems, and Figure 6.B shows six pole-zero plots.

For each of the systems in Figure 6.A, suggest which of the pole-zero plots could have produced that response in open loop. (There is only one viable option per plot.)

Total: 6 marks

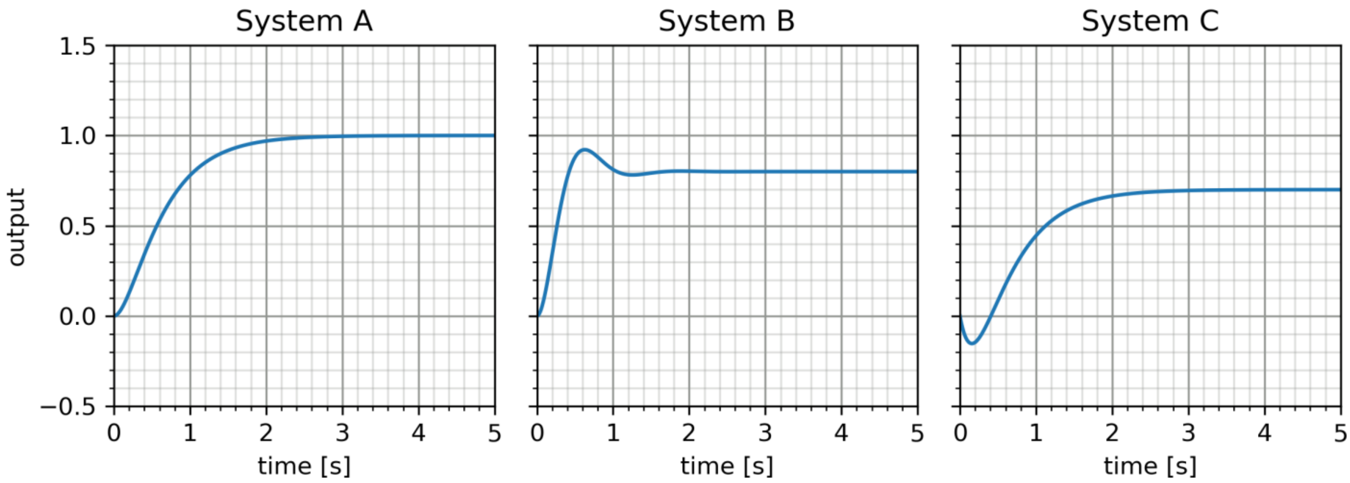


Figure 6.A – unit step responses

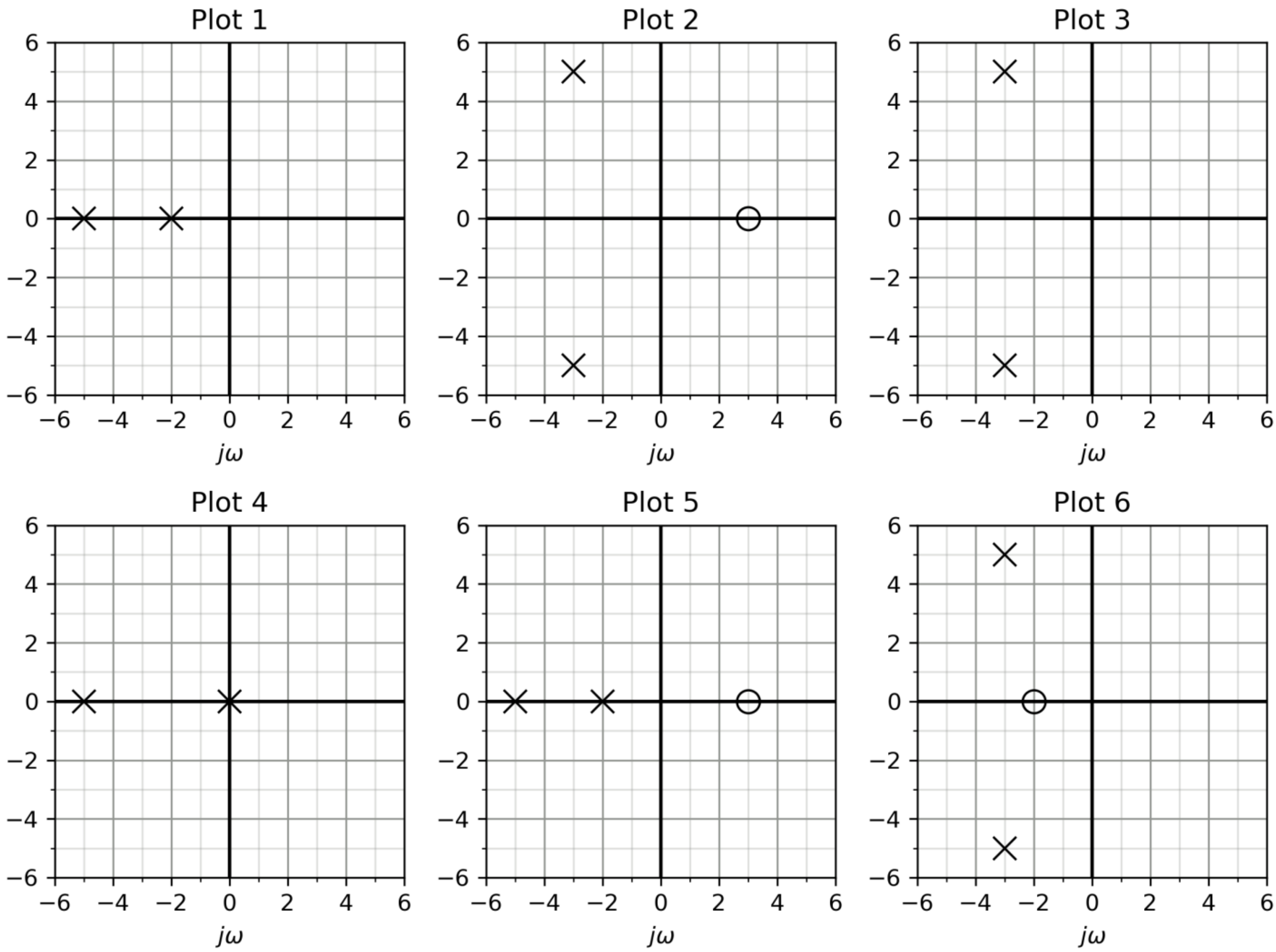


Figure 6.B – pole and zero plots

Solution:

System A – plot 1: This response has no overshoot, suggesting that the poles are real. Plot 1 and plot 4 show real poles, but one of the poles is at the origin in 4, which would lead to a ramp response in open loop, so the correct plot must be 1.

System B – plot 3: This response overshoots the final value, indicating that the poles are complex. Plots 2, 3 and 6 have complex pole pairs, but plots 2 and 6 include zeros, which would affect the initial value of the response (if they are in the left-half plane) or cause nonminimum phase behaviour (if they are in the right half), so 3, without zeros, must be the solution.

System C – plot 5: This response moves in the opposite direction to the final value initially, indicating the presence of a right-half plane zero (i.e. the system is nonminimum phase.) Plots 2 and 4 include right-half plane zeros, but the response does not overshoot, so it is a better match for the real poles in 5.

Question 7

Consider the system $G(s) = \frac{1}{(s+2)(s+4)}$

A) The system is connected in closed loop.
   i) [3 marks] What is the minimum possible closed-loop settling time (to within 2% of the final value) for this system? You may sketch a rough root locus plot to support your answer.

Solution: The breakaway point of the root locus is at -3, between the two poles. At small gains, where the closed-loop poles are real, one of them will always be closer to the origin than -3, and therefore slower, so the response will be fastest when the response is critically- or underdamped and both have a real part of -3. This corresponds to a time constant of $\tau = \frac{1}{3}$ and settling time to within 2% of $4\tau = 1.33$ seconds.

   ii) [3 marks] Calculate the gain required from a proportional controller to achieve this settling time and a damping ratio of 0.7.

Solution: The natural frequency can be calculated from the real part of the pole and damping ratio to be $\omega_n = \frac{3}{0.7} = 4.29$ rad/s. The imaginary part of the pole will be $\sqrt{4.29^2 - 3^2} = 3.06$

The target closed-loop pole pair is therefore $s = 3 \pm j3.06$

The gain required to produce these poles is $K = \frac{1}{|G(3+j3.06)|} = |(3 + j3.06 + 2)(3 + j3.06 + 4)| = 10.4$

B) The closed-loop system is required to track a position input with 95% accuracy.
   i) [2 marks] Calculate the error constant for a unit step input when the proportional controller you designed in part A.ii is included in the system.

Solution: The open-loop DC gain will be $K_{dc} = \frac{10.4}{(2)(4)} = 1.72$, so the closed-loop error constant will be $\frac{1}{1+K_{dc}} = 0.37$

   ii) [4 marks] Design a lag compensator to achieve the required accuracy. The combined system should have the same damping characteristics as in part A.ii.

The open-loop gain must be at least 19 to reduce the position error to < 5%, meaning that the lag compensator must increase the gain by a factor of at least 11.04.

The zero must therefore be at least 11.04 times further from the origin than the pole is.

The pole and zero should be placed close to the origin. Two rules of thumb were suggested in class:

1) the zero should be 10 – 50 x closer to the origin than the closest system pole
2) the lag compensator must have an angular contribution of less than 2 degrees at the target closed-loop pole.

   iii) [1 marks] Do you expect that the lag-compensated system will settle in more or less time than the value you worked out in part A.i?

Solution: more time, as the lag compensator adds a very slow pole.

C) [2 marks] A colleague suggests using a controller of the form $C(s) = \frac{K}{s}$ to satisfy the accuracy requirement instead. Briefly discuss the advantages and disadvantages of this idea.

Solution: The pros and cons of using the integrator are as follows:

Pros: perfect tracking accuracy

Cons: this is an ideal compensator, and therefore cannot be built using only passive components.

Total: 15 marks